

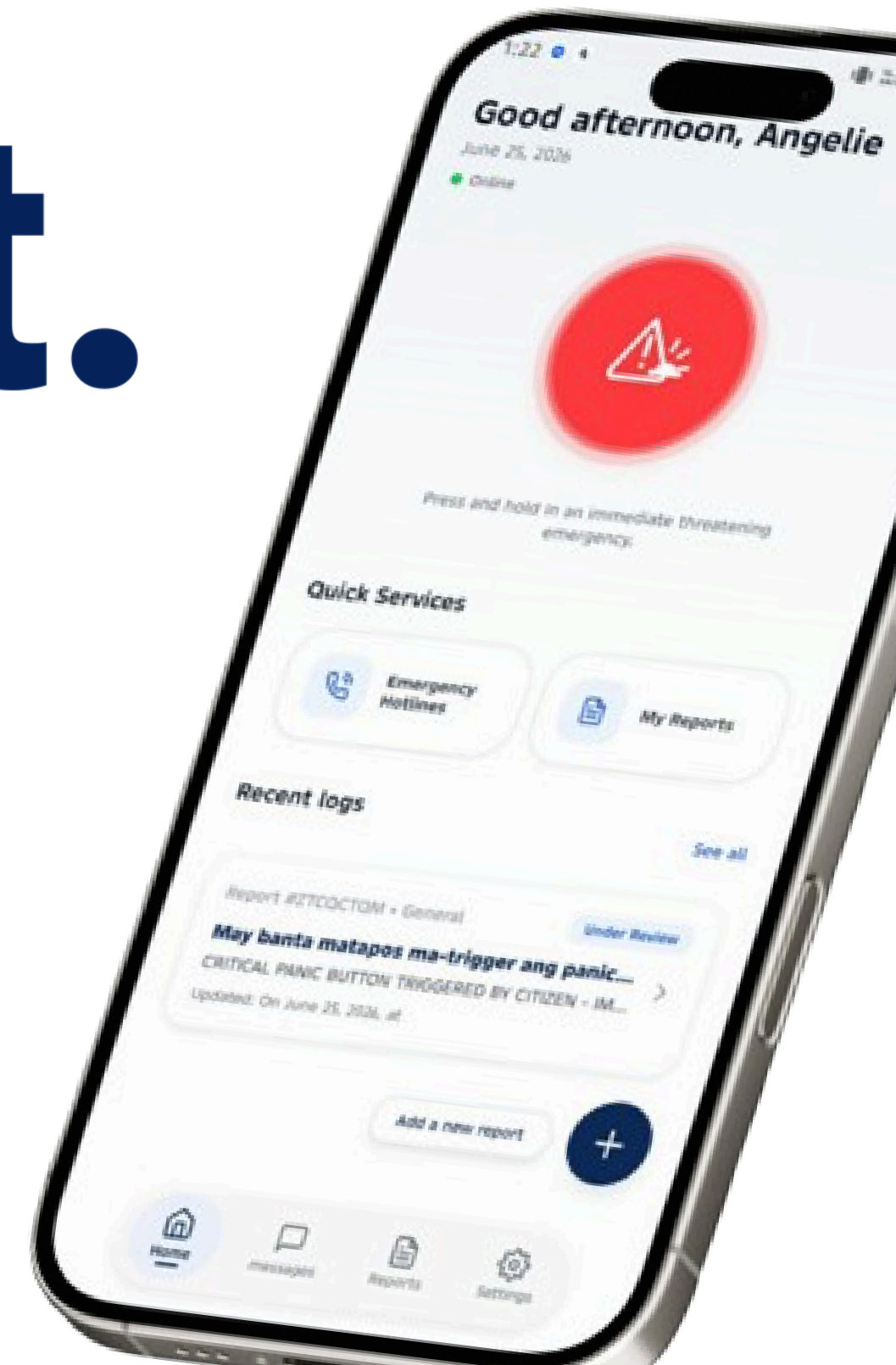
CAPSTONE 1

BrgyAlert.

A Responsive Incident Reporting and Analytics
Platform for Barangay Lepa

Capstone Team:

Project Manager:	Angelie Quirante
Lead Programmer:	Thyla Faye Quinto
UI/UX Designer:	Angelie Quirante
System Analyst/Specialist:	Christian Joeie Penoliar



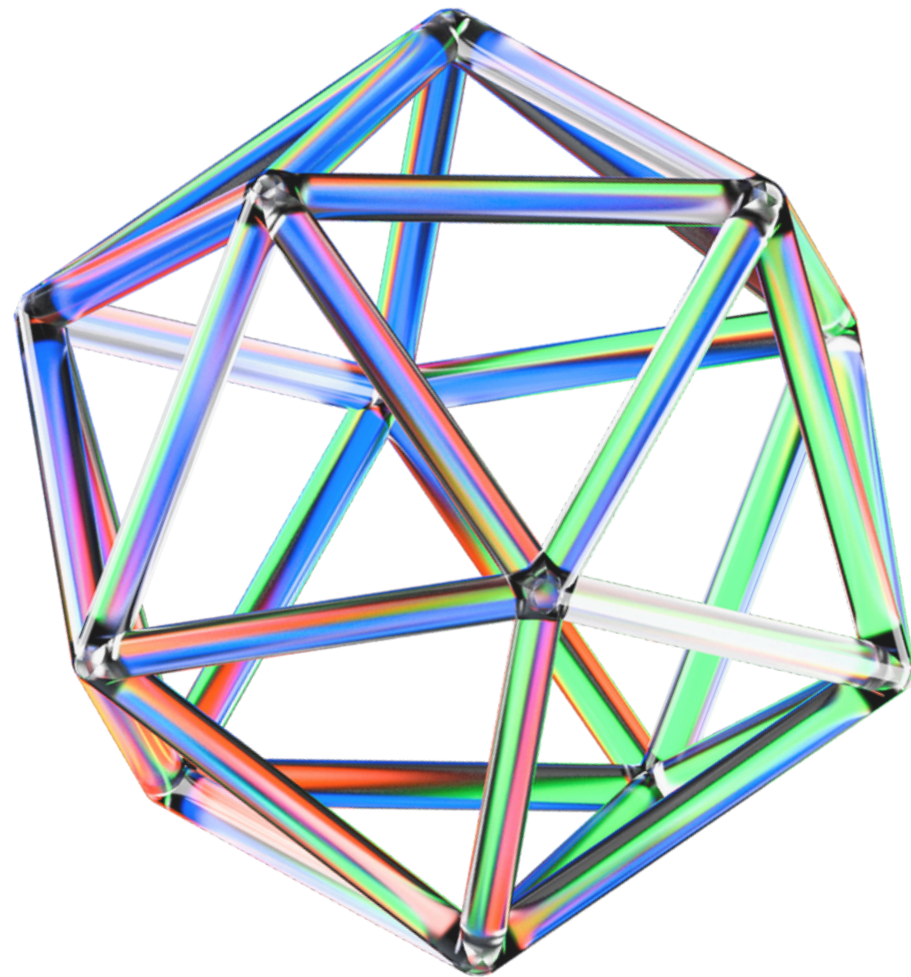
BACKGROUND OF THE STUDY



The Socio-Technical Void

While Philippine barangays are legally mandated by DILG MC No. 2017-144 to operate active Violence Against Women (VAW) Desks, Barangay Lepa still suffers from an "analog lag" caused by manual, paper-based incident logging. This manual system creates a critical socio-technical void marked by fragmented data, slow processing times for life-saving Barangay Protection Orders (BPOs), and desk-bound limitations for on-the-move responders. Furthermore, the lack of private reporting channels reinforces a dangerous "culture of silence" among victims due to social stigma and fear of retaliation. To bridge this gap, this study develops BrgyAlert, a mobile-first platform that integrates a secure cloud backend with an AI-driven risk stratification engine. By leveraging Natural Language Processing (NLP) to automatically triage situational urgency into Low, Medium, or High-risk levels, BrgyAlert eliminates administrative bias, protects victim confidentiality, and transitions grassroots local governance into a proactive, data-driven safety ecosystem.

THEORETICAL FRAMEWORK



Socio-Technical Systems

Aligns administrative human workflows with algorithmic triage tools, ensuring responders work in sync with technical platform alerts.



Technology Acceptance

Derived from TAM; emphasizes Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) to break local cultures of silence.



UTAUT Framework

Addresses local infrastructure assets (network conditions, device availability) ensuring organic institutional adoption.

STATEMENT OF THE PROBLEM

1 What are the functional, non-functional, and security requirements of the BrgyAlert mobile application based on the current field challenges of the Barangay Lepa Violence Against Women (VAW)Desk?

1.1. What are the specific user requirements for a discreet and responsive mobile reporting interface for victims?

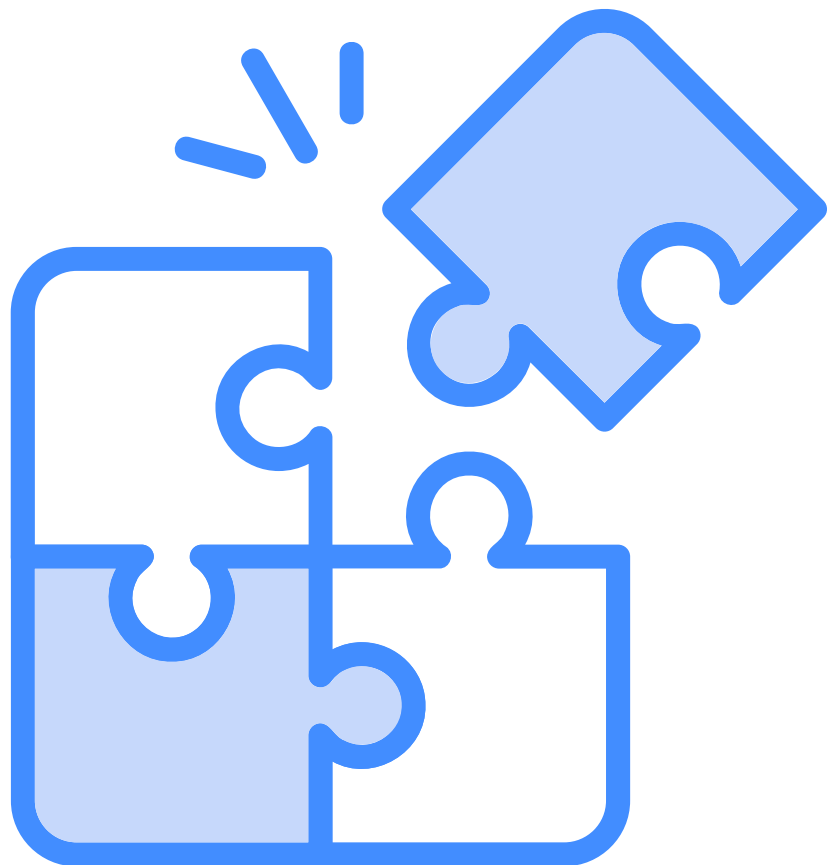
1.2. What administrative prerequisites must barangay officials meet in order to use an automated analytical dashboard?

1.3. What encryption and data security procedures are necessary to guarantee the confidentiality and integrity of incident reports?

2 How can local incident reporting and data processing be optimized in the design and development of the BrgyAlert system?

2.1. How will the Artificial Intelligence (AI) engine be designed to classify situational risk urgency and analyze behavioral data inputs?

2.2. How will the secure database of the application be set up to guarantee data integrity and stop illegal access?



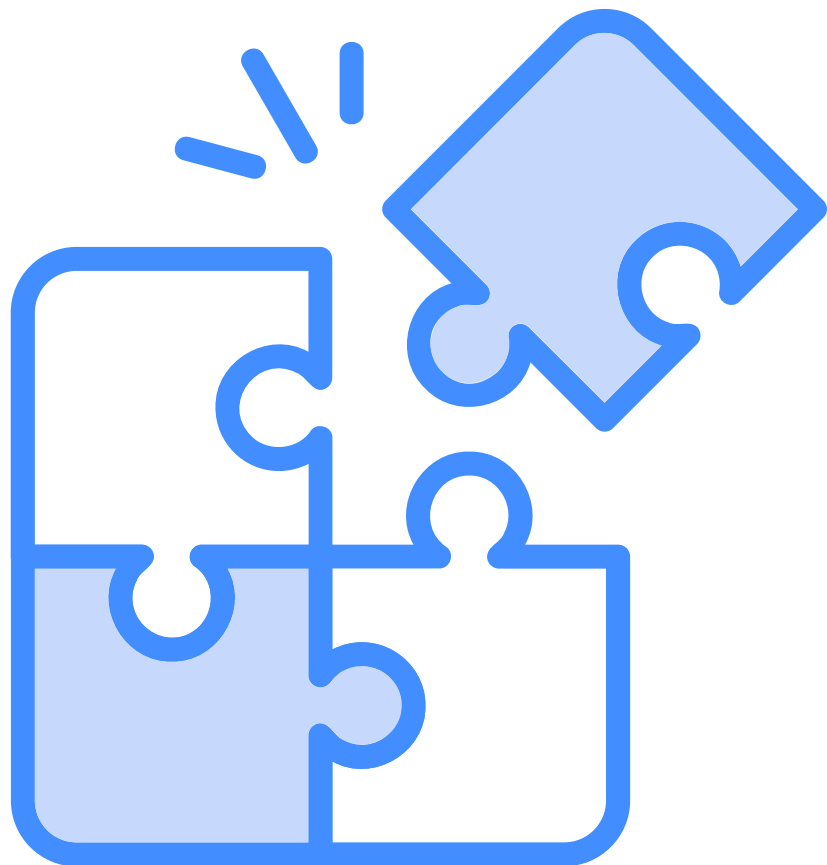
STATEMENT OF THE PROBLEM

3 What is the level of acceptability of the BrgyAlert application among its primary stakeholders in terms of TAM variables?

3.1. What is the mobile interface's perceived ease of use (PEOU) for barangay employees and community reporters?

3.2. What is the AI risk stratification engine's perceived usefulness (PU) in terms of expediting protective order workflows and improving community safety?

4 How does the developed BrgyAlert mobile application improve the efficiency of incident documentation and the processing time of Barangay Protection Orders (BPOs) compared to the traditional manual blotter system?



CONCEPTUAL FRAMEWORK

INPUT

Citizen incident inputs (TAM), Mobile network resources, DILG mandates, and RA 10173 legal compliance rules.

PROCESS

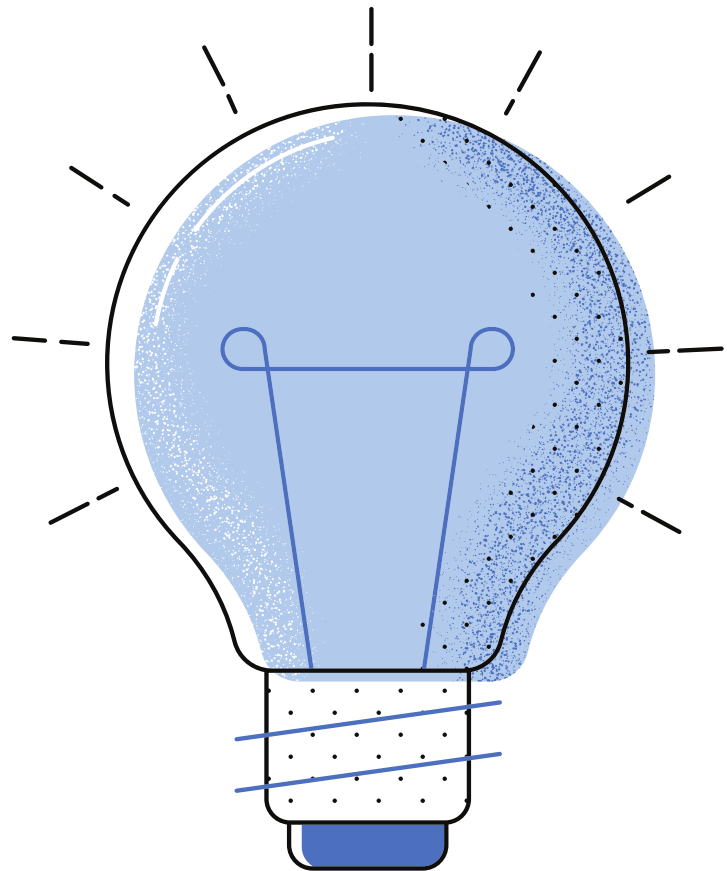
Iterative Agile engineering, User Interface Wireframing, Cloud AI Risk Stratification engine model tuning.

OUTPUT

Secure BrgyAlert app, centralized response analytics dashboard, and fast-tracked Protection Order workflows.

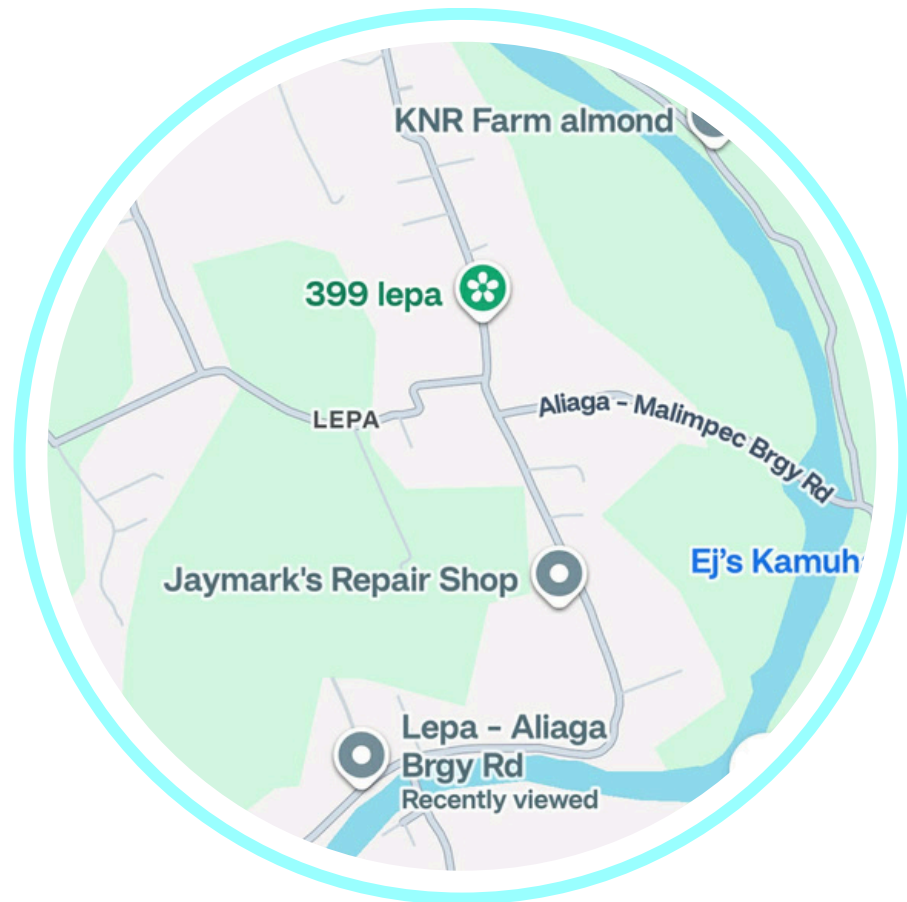


OBJECTIVE OF THE STUDY



- Identify Requirements: Determine the functional, non-functional, and security specifications needed by victims and VAW Desk personnel.
- Develop the System: Build a mobile-first application using React Native and a secure Firebase cloud infrastructure.
- Implement AI Triage: Integrate a cloud-assisted AI engine using Google Gemini to automatically classify incident urgency into Low, Medium, or High risk.
- Evaluate Acceptability: Measure stakeholder adoption based on the Technology Acceptance Model (TAM) variables: Perceived Ease of Use and Perceived Usefulness.
- Assess Efficiency: Compare the digital platform against the manual paper blotter to measure improvements in documentation speed and Barangay Protection Order (BPO) issuance times.

SCOPE AND DELIMITATION



In-Scope Infrastructure

- Tailored specifically for the micro-governance of Barangay Lepa.
- Cross-platform React Native client app for residents.
- Cloud-assisted Google Gemini AI API integration for automatic incident priority flagging.
- Firebase cloud integration for secure, immediate database synchronization.

Platform Limitations

- Geofenced strictly within the administrative borders of Barangay Lepa.
- Independent database structure isolated from the national PNP-CIRAS system.
- Basic offline fallback via SMS; AI categorization requires active network connectivity.

SIGNIFICANCE OF THE STUDY



Vulnerable Victims

Provides a safe, discreet, and private avenue to seek assistance without fear of domestic retaliation.



VAW Desk Officers

Automates manual registry lookup, shortening administrative delays when filing Protection Orders (BPO).



Barangay Council

Offers spatial crime trend mapping to direct patrols (Tanods) to active hotspots intelligently.

REVIEW OF RELATED LITERATURE AND STUDY



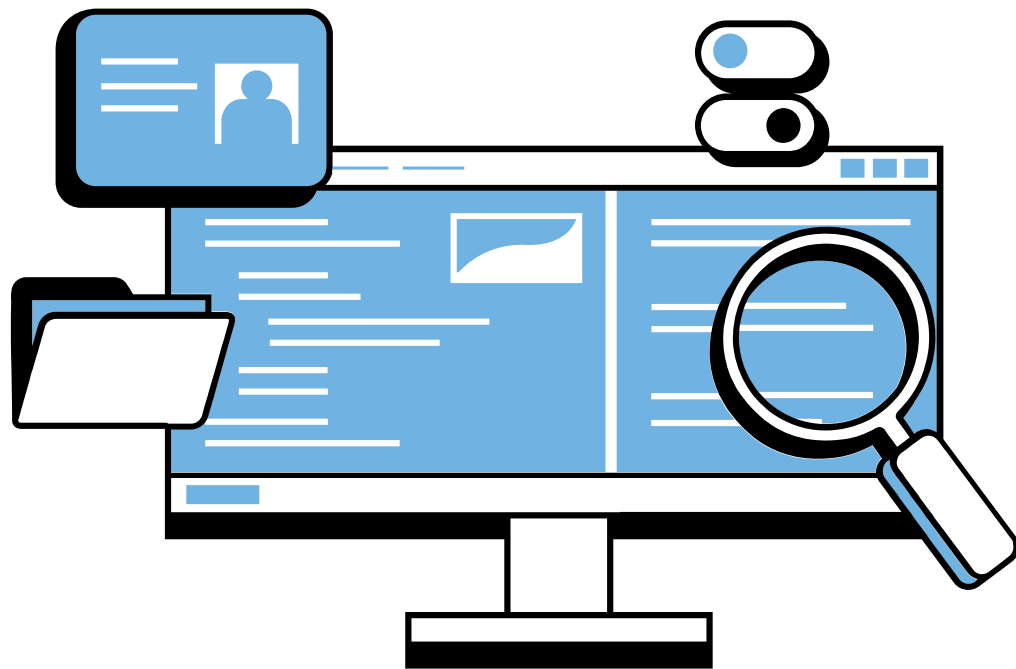
Adiyono et al. (2024), Ssaho (2024), Chen et al. (2021)	Highlights that municipal apps must have high structural simplicity to avoid cognitive overload for local users, emphasizing that ease of use drives technology adoption.
Novitzky et al. (2023), Anderson & Miller (2022), Lee & Kim (2023)	Novitzky et al. (2023), Anderson & Miller (2022), Lee & Kim (2023) Proves that machine learning and NLP can accurately analyze linguistic markers to detect domestic violence severity and remove subjective human bias during triage.
Levy & Snell (2020), Braun & Schmidt (2023), RA 10173 (Data Privacy Act)	Emphasizes that systems handling sensitive abuse data require high-level encryption and secure cloud architectures to maintain a forensic trail and prevent "digital control" by abusers.
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Santiago & Dela Cruz (2022), Weger & Yeazitzis (2023), DILG MC 2017-144	Identifies a "socio-technical void" where manual, paper-based barangay blotters cause fragmented records and delay life-saving protective workflows.

SYNTHESIS OF THE REVIEW



The literature confirms a persistent analog lag at the grassroots. While legal mandates dictate protection, manual processes leave records insecure and delay response. BrgyAlert bridges this socio-technical void by blending localized mobile access with cloud-assisted AI triage.

RESEARCH DESIGN



Descriptive-Evaluative Model

The study utilizes a Descriptive-Evaluative approach to collect baseline pain points from responders, while testing system acceptability among residents.

Quantitative evaluations are analyzed via a structured pre-survey survey with a Dichotomous Scale.

Compliance is preserved strictly under RA 10173, protecting all resident profiles during tests.

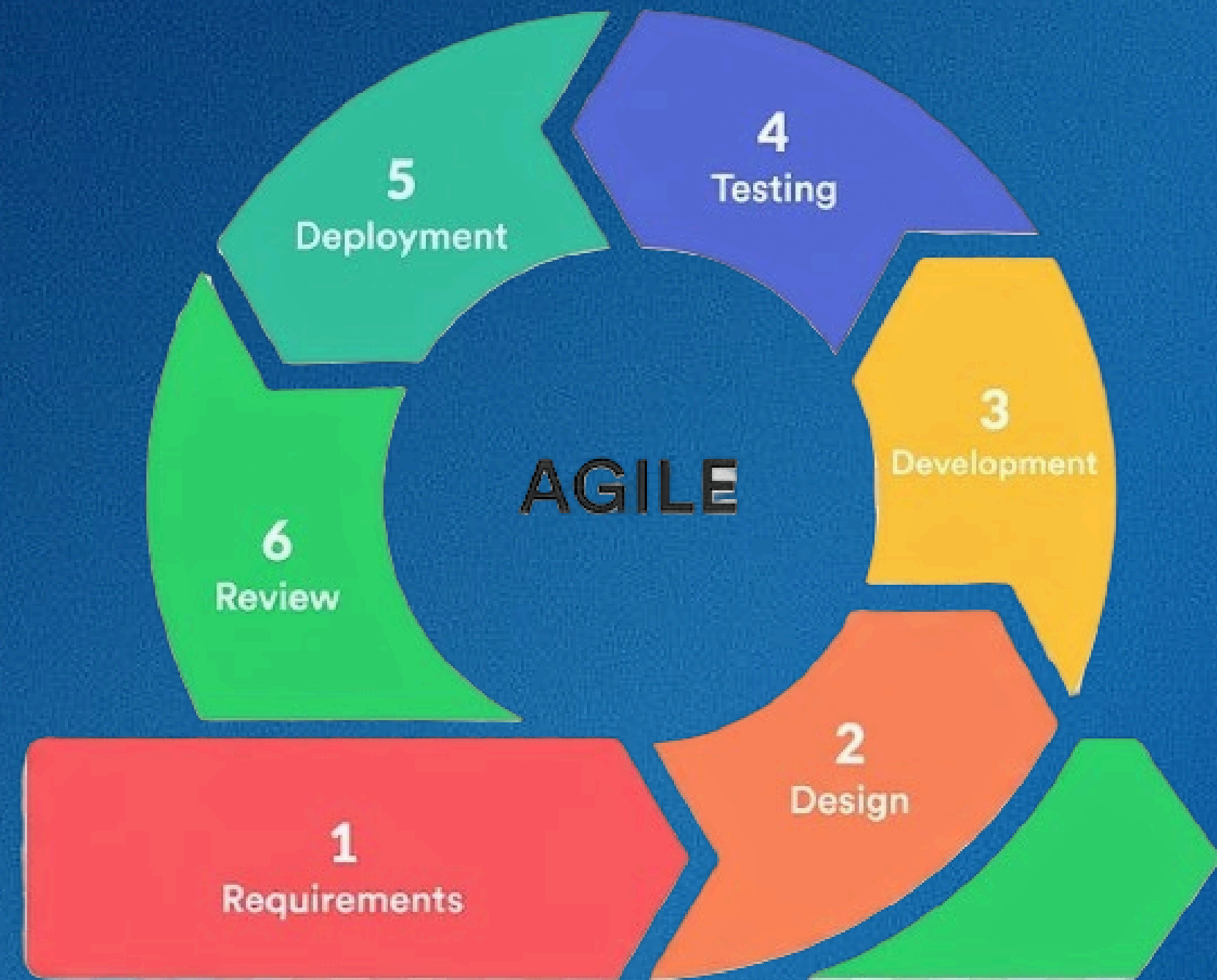
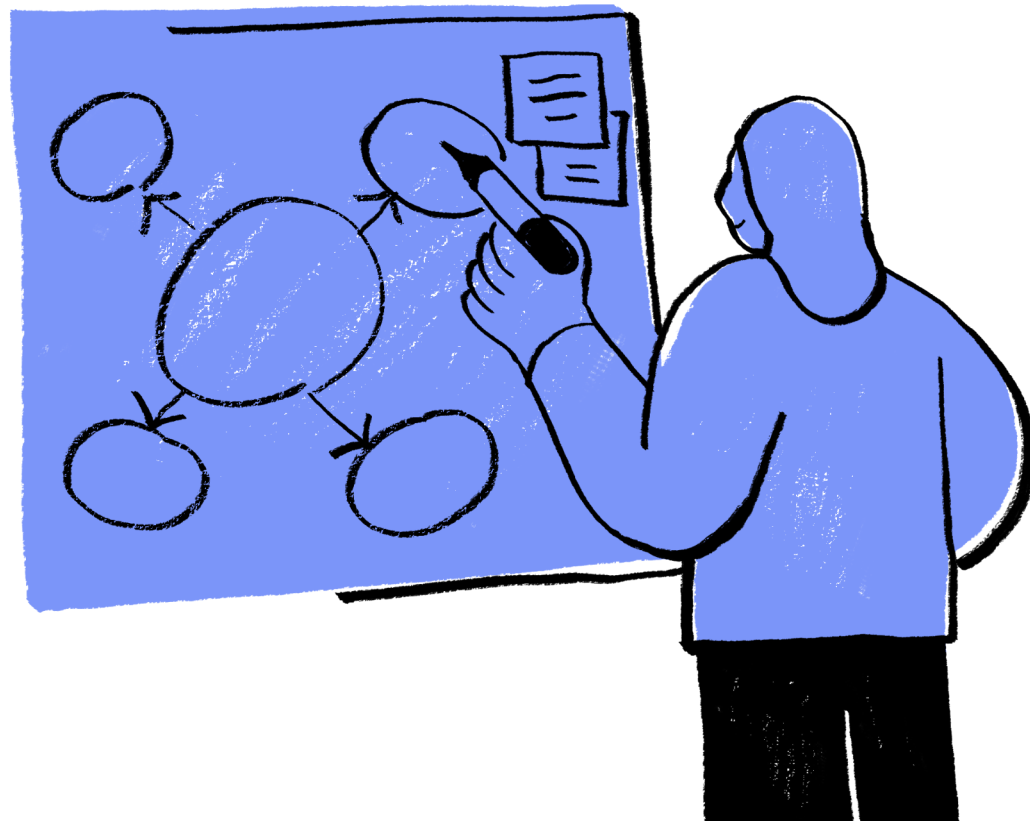
Descriptive Statistical Tools

The researchers calculate simple percentages to analyze response rates for each item:

$$P = \frac{f}{n} \times 100$$

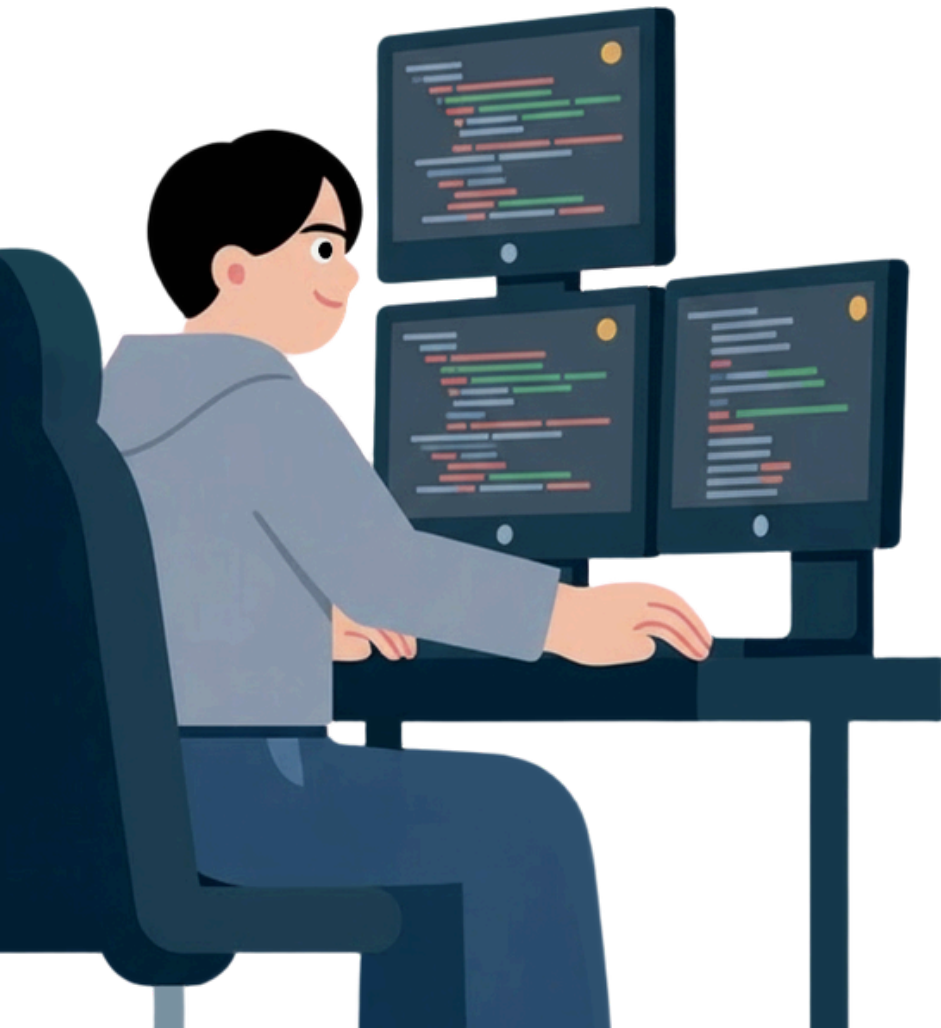
Where: **P** = Percentage, **f** = frequency, and **n** = total sample.

AGILE METHODOLOGY

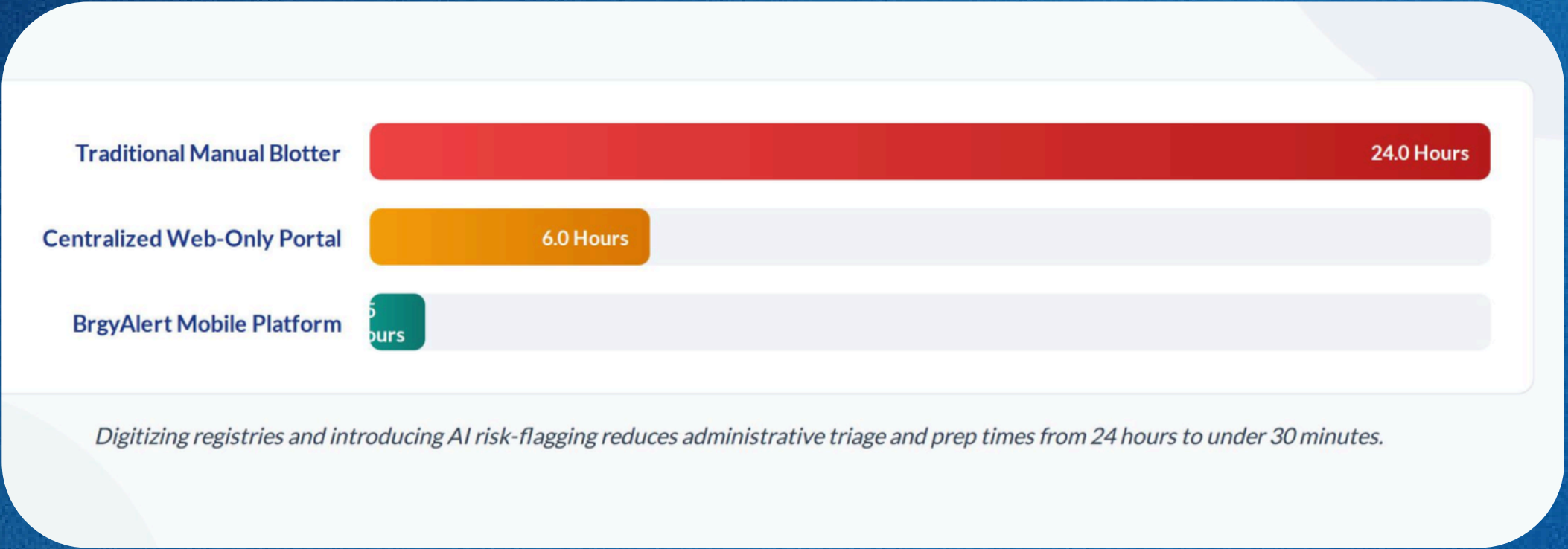
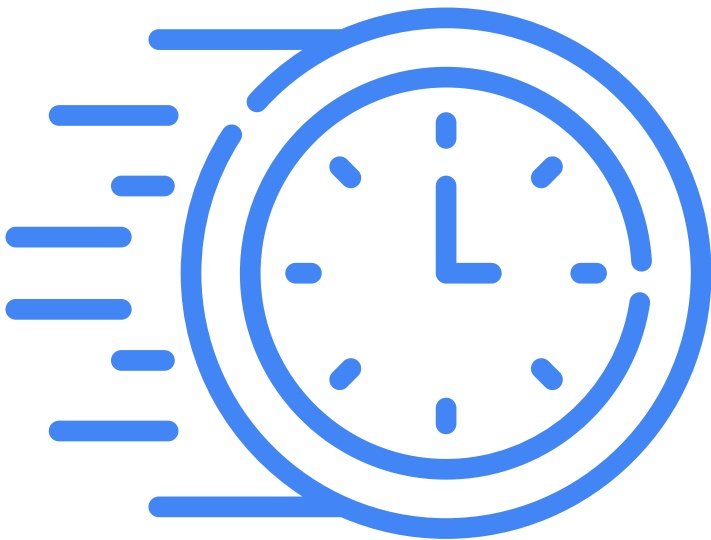


TECHNICAL & DATA REQUIREMENTS

Infrastructure Class	Core Components Specified	Functional System Role
Software Architecture	React Native, Expo go, Firebase Database, Google Gemini API	Enables active synchronization, multiplatform access, and intelligent text classification.
Hardware Deployment	Android & IOS Mobile Devices,	Provides on-the-go logging for responders and centralized oversight for the VAW Desk.
Data Security	AES-256 Cloud encryption layers, secure hashing protocols	Preserves forensic logs and locks down delicate victim profiles to prevent leaks.



BPO ISSUANCE PROCESSING SPEED



THANK YOU!